NLP Resume Matching System

Resume & Job Description Matching Pipeline

Automated text extraction and vector similarity evaluation between candidate resume and target role specification.

Core Libraries	Algorithm Model	Notebook Output
sklearn, PyPDF2	CountVectorizer & Cosine Similarity	Match Score: 59.15 %

CORE IMPLEMENTATION

Python Matching Algorithm

```
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics.pairwise import cosine_similarity
import PyPDF2

def read_pdf(file_path):
    text = ""
    with open(file_path, "rb") as file:
        reader = PyPDF2.PdfReader(file)
        for page in reader.pages:
            text += page.extract_text()
    return text

def match_score(resume, job_desc):
    texts = [resume.lower(), job_desc.lower()]
    vectorizer = CountVectorizer().fit_transform(texts)
    vectors = vectorizer.toarray()
    score = cosine_similarity([vectors[0]], [vectors[1]])[0][0]
    return round(score * 100, 2)
```

EXECUTION OUTPUT

Calculated Match Result

Cosine Similarity Output

59.15 %

Computed match score from notebook stdout.

INPUT RESUME

`resume edirick kpmg audit.pdf`

VECTOR PIPELINE

Texts lowercased, vectorised into 2D array, and measured using pairwise cosine similarity.

Experience and Qualifications

- Bachelor's Degree in Finance, Business or Mathematics/Statistics.
- 1–2 of experience in a data processing/data administrator or analyst role.
- Fluency in English both oral and written.
- Basic understanding of financial markets.
- Previous database administration experience.
- Advanced skills in Excel & Good understanding of Microsoft Office Suite and internet navigation.

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Additional Skills & Policy Text

Additional Skills & Requirements

- Building a high performance and inclusive team culture.
- High standard of focus, accuracy and attention to detail.
- Strong analytical and problem-solving skills.
- Ability to use initiative in identifying issues and recommending actions.
- Proficiency in Spanish will be considered an advantage.

Equal Opportunity Statement

> *"We believe in equality of opportunity and welcome applications from individuals, regardless of age, ethnicity, disability, gender identity, sex, sexual orientation, socio-economic background, religion and/or belief, or any other classification protected by applicable laws."*

NOTEBOOK SUMMARY

Notebook Pipeline Breakdown

1. PyPDF2 Extraction

Iterates through each PDF page to concatenate raw textual content.

2. CountVectorizer

Converts lowercased resume and job description text into token count vectors.

3. Cosine Score

Calculates vector angle yielding the resulting match of 59.15%.